

WILP, BITS Pilani

Session 15: Gaussian Mixture Models & Practice Problems

Course: Introduction to Statistical Methods (ISM) | Instructor: Dr. YVK Ravi Kumar

1 Gaussian Mixture Models (GMM)

Model complex multimodal data by combining several weighted Gaussian distributions.

A Gaussian Mixture Model represents an overall population probability density as a linear combination of K underlying Gaussian components:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x | \mu_k, \Sigma_k) \quad (1)$$

where π_k are mixing coefficients satisfying $\sum_{k=1}^K \pi_k = 1$ and $\pi_k \geq 0$.

Everyday Picture & Intuition

Suppose share prices of Company A average \$90 and Company B average \$100. When combined into one pool, the price distribution shows two distinct peaks. GMM models this combined density as a weighted blend of two bell curves.

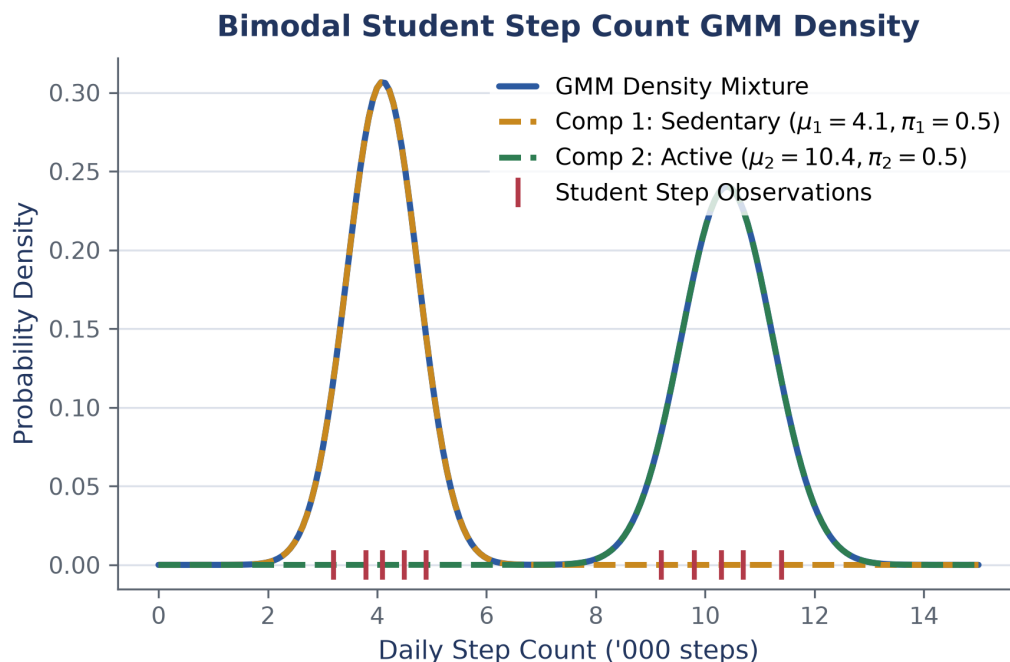


Figure 1: Bimodal student step count density modeled using a 2-component Gaussian Mixture Model.

2 The Expectation-Maximization (EM) Algorithm

Iteratively estimate component parameters and soft cluster responsibilities.

Because cluster assignments are unobserved latent variables $z_k \in \{0, 1\}$, parameter likelihood cannot be solved in closed form. The Expectation-Maximization (EM) algorithm alternates between two steps until convergence:

2.1 E-Step (Expectation)

Compute posterior responsibilities $\gamma(z_k)$ using Bayes' theorem:

$$\gamma(z_k) = P(z_k = 1 \mid x) = \frac{\pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x \mid \mu_j, \Sigma_j)} \quad (2)$$

2.2 M-Step (Maximization)

Re-estimate component parameters using current responsibilities $N_k = \sum_{i=1}^N \gamma(z_{ik})$:

$$\mu_k^{\text{new}} = \frac{1}{N_k} \sum_{i=1}^N \gamma(z_{ik}) x_i \quad (3)$$

$$\Sigma_k^{\text{new}} = \frac{1}{N_k} \sum_{i=1}^N \gamma(z_{ik}) (x_i - \mu_k^{\text{new}})(x_i - \mu_k^{\text{new}})^T \quad (4)$$

$$\pi_k^{\text{new}} = \frac{N_k}{N} \quad (5)$$

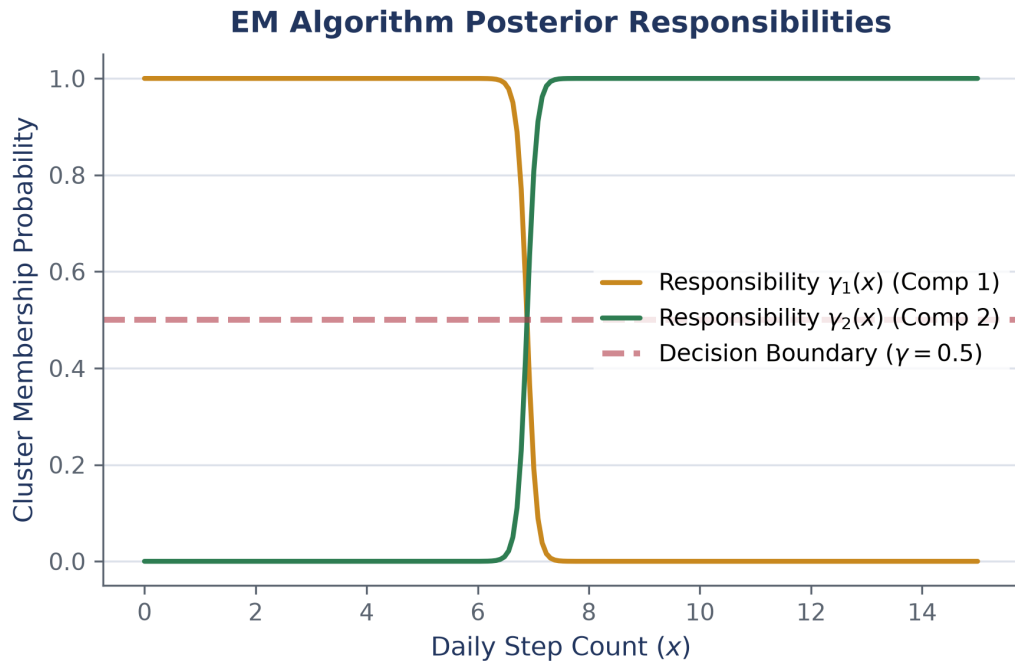


Figure 2: Soft cluster responsibilities $\gamma_k(x)$ assigning probabilities across data points.

Fully Worked Example

A wellness app logs step counts (in thousands) of 10 students: {3.2, 3.8, 4.1, 4.5, 4.9, 9.2, 9.8, 10.3, 10.7, 11.4}. Component 1 consists of the first 5 values (sedentary); Component 2 consists of the last 5 values (active).

Step 1: Calculate mixing coefficients:

$$\pi_1 = \frac{5}{10} = 0.50, \quad \pi_2 = \frac{5}{10} = 0.50 \quad (6)$$

Step 2: Calculate component means:

$$\begin{aligned} \mu_1 &= \frac{3.2 + 3.8 + 4.1 + 4.5 + 4.9}{5} = \frac{20.5}{5} = 4.10 \\ \mu_2 &= \frac{9.2 + 9.8 + 10.3 + 10.7 + 11.4}{5} = \frac{51.4}{5} = 10.28 \end{aligned}$$

Step 3: Calculate component variances $s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$:

$$\begin{aligned} s_1^2 &= \frac{(-0.9)^2 + (-0.3)^2 + 0^2 + 0.4^2 + 0.8^2}{4} = \frac{1.70}{4} = 0.425 \\ s_2^2 &= \frac{(-1.08)^2 + (-0.48)^2 + 0.02^2 + 0.42^2 + 1.12^2}{4} = \frac{2.828}{4} = 0.707 \end{aligned}$$

Step 4: Formulate total GMM PDF:

$$p(x) = 0.50\mathcal{N}(x \mid 4.10, 0.425) + 0.50\mathcal{N}(x \mid 10.28, 0.707) \quad (7)$$

3 Proportions & Chi-Square Independence Tests

Test differences in proportions and independence in contingency tables.

To test proportion differences across samples or test independence in a $2 \times k$ contingency table, we compute expected frequencies $E_{ij} = \frac{R_i C_j}{N}$ and test statistic $\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$.

Fully Worked Example

Employees from Agency 1 (100 total: 67 For, 33 Against) and Agency 2 (150 total: 84 For, 66 Against) were surveyed about a pension plan. Test for equal proportions at $\alpha = 0.01$.

Step 1: Overall pooled proportion favoring pension: $\hat{p} = \frac{67+84}{100+150} = \frac{151}{250} = 0.604$.

Step 2: Expected frequencies: $E_{11} = 100 \times 0.604 = 60.4$, $E_{21} = 150 \times 0.604 = 90.6$.

Step 3: Compute Chi-Square statistic:

$$\begin{aligned} \chi^2 &= \frac{(67 - 60.4)^2}{60.4} + \frac{(33 - 39.6)^2}{39.6} + \frac{(84 - 90.6)^2}{90.6} + \frac{(66 - 59.4)^2}{59.4} \\ &= 0.721 + 1.100 + 0.481 + 0.733 = 3.035 \end{aligned}$$

Step 4: Critical value at $\alpha = 0.01$, $df = 1$ is 6.635. Since $\chi^2 = 3.035 < 6.635$, fail to reject H_0 . Proportions favoring the plan do not differ significantly between agencies.

Watch Out! Pitfalls

In GMM parameter estimation, initialization matters! Bad initial guesses for component means can cause the EM algorithm to get stuck in local optima.

Key Takeaways

- GMM models complex distributions as linear combinations of K Gaussians with mixing weights π_k .
- EM algorithm alternates between E-step (computing posterior responsibilities) and M-step (updating means and covariances).
- Chi-square contingency tests compare observed vs expected frequencies across categorical proportions.